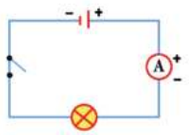
- ❖ Parallel Circuit — A circuit in which there are multiple paths for electricity to flow. Each load connected in a separate path receives the full circuit voltage, and the total circuit current is equal to the sum of the individual branch currents.
- ❖ Series Circuit — A circuit in which there is only one path for electricity to flow. All of the current in the circuit must flow through all of the loads.
- ❖ Short Circuit — When one part of an electric circuit comes in contact with another part of the same circuit, diverting the flow of current from its desired path.
- ❖ One unit of coulomb is charge of approximately 6.242×10^{18} protons or electrons.
- ❖ The potential difference between any two points is the amount of energy needed to move one unit of electric charge from one point to the other.
- ❖ Electrical conductivity or specific conductance is the measures a material's ability to conduct an electric current
- ❖ Electrical resistivity is the property of a material that quantifies how strongly that material opposes the flow of electric current.
- ❖ The sources which produce the small amount of electricity for shorter periods of time is called as electric cell or electro chemical cells.
- ❖ Electrolytes : A substance that dissociates into ions in solution and acquires the capacity to conduct electricity. Sodium, potassium, chloride, calcium, and phosphate are examples of electrolytes.

EVALUATION

I. Choose the correct answers

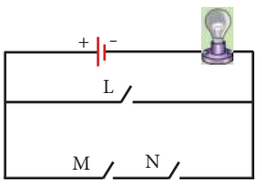
ETCL9

1. In the circuit diagram below, 10 units of electric charge move past point x every second .What is the current in the circuit?



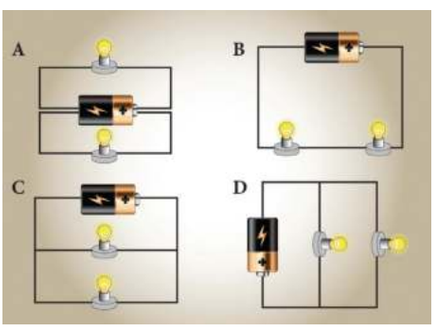
   a) 10 A b) 1 A
   c) 10 V d) 1 V

2. In the circuit shown, which switches (L,M or N) must be closed to light up the bulb?



   a) switch L only
   b) switch M only
   c) Switch M and N only
   d) either switch L or switches M and N

3. Small amounts of electrical current are measured in milliampere (mA). How many milliampere are there in 0.25 A ?
   a) 2.5 mA b) 25 mA
   c) 250 mA d) 2500 mA
4. In which of the following circuits are the bulb connected in series?



II. Fill in the blanks.

1. The direction of conventional current is _____ to electron flow.

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